

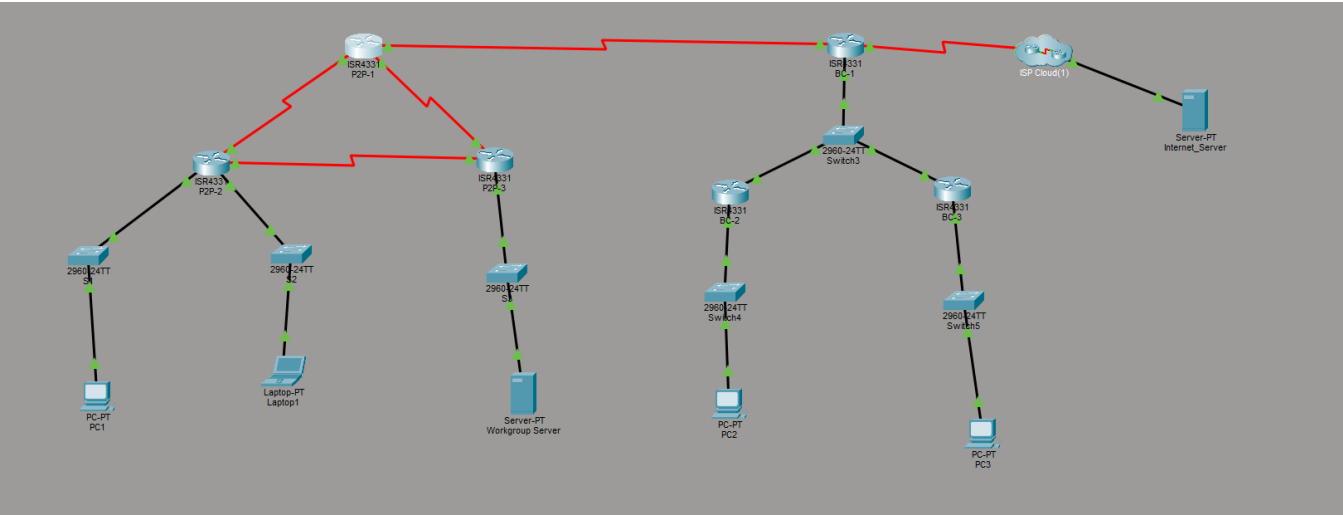
LAB 13

นายจิรภัทร แก้วดี 673380577-9 Sec 2

Device	Interface	IP Address / Prefix
P2P-1	S0/1/0	10.0.0.1/30
	S0/1/1	10.0.0.9/30
	S0/2/0	10.0.0.13/30
P2P-2	S0/1/0	10.0.0.2/30
	S0/1/1	10.0.0.5/30
	G0/0/0	192.168.1.1/24
	G0/0/1	192.168.2.1/24
P2P-3	S0/1/0	10.0.0.6/30
	S0/1/1	10.0.0.10/30
	G0/0/0	192.168.3.1/28
BC-1	S0/1/0	10.0.0.14/30
	S0/1/1	64.0.100.2/30
	G0/0/0	10.0.1.1/29
BC-2	G0/0/0	192.168.4.1/30
	G0/0/1	10.0.1.2/29
BC-3	G0/0/0	192.168.5.1/24
	G0/0/1	10.0.1.3/29
Internet Server	NIC	203.0.113.100/24
PC 1	NIC	192.168.1.10/24

Device	Interface	IP Address / Prefix
Laptop 1	NIC	192.168.2.20/24
Workgroup Server	NIC	192.168.3.14/28
PC 2	NIC	192.168.4.40/24
PC 3	NIC	192.168.5.50/24

topology



การตั้งค่าอุปกรณ์แต่ละเครื่อง

```
P2P-1#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
P2P-1(config)#interface Serial0/1/0
P2P-1(config-if)# ip address 10.0.0.1 255.255.255.252
P2P-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
P2P-1(config-if)#interface Serial0/1/1
P2P-1(config-if)# ip address 10.0.0.9 255.255.255.252
P2P-1(config-if)# ip ospf cost 50
P2P-1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
P2P-1(config-if)#interface Serial0/2/0
P2P-1(config-if)# ip address 10.0.0.13 255.255.255.252
P2P-1(config-if)# ip ospf hello-interval 20
P2P-1(config-if)# ip ospf dead-interval 80
P2P-1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
P2P-1(config-if)#router ospf 10
P2P-1(config-router)# auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
P2P-1(config-router)# network 10.0.0.0 0.0.0.3 area 0
P2P-1(config-router)# network 10.0.0.8 0.0.0.3 area 0
P2P-1(config-router)# network 10.0.0.12 0.0.0.3 area 0
P2P-1(config-router)#end
P2P-1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
P2P-1#
```

แสดงคำสั่ง config ทั้งหมดของ P2P-1 ได้แก่ การกำหนด IP Address บน Serial0/1/0, Serial0/1/1 (พร้อม cost 50), Serial0/2/0 (พร้อม hello/dead-interval 20/80) และการเปิดใช้งาน OSPF ด้วย network statement ครอบคลุม 3 subnet ตาม requirement ของ Headquarters

```

Router#
Router#
Router#
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/1/0
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config-if)#interface Serial0/1/1
Router(config-if)# ip address 10.0.0.5 255.255.255.252
Router(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
Router(config-if)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#router ospf 10
Router(config-router)# passive-interface GigabitEthernet0/0/0
Router(config-router)# passive-interface GigabitEthernet0/0/1
Router(config-router)# auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router(config-router)# network 10.0.0.4 0.0.0.3 area 0
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#end
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
00:20:24: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING to FULL,
Loading Done
|

```

แสดงคำสั่ง config ของ P2P-2 ได้แก่ IP Address บน Serial และ Gigabit interface ทั้งหมด, การตั้ง passive-interface บนพอร์ต LAN (G0/0/0, G0/0/1), และ network statement ครอบคลุม subnet โดยด้านล่างมี log ยืนยันว่า OSPF adjacency กับ P2P-1 เปลี่ยนสถานะเป็น **FULL** สำเร็จ (Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING to FULL)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname P2P-3
P2P-3(config)#interface Serial0/1/0
P2P-3(config-if)# ip address 10.0.0.6 255.255.255.252
P2P-3(config-if)# no shutdown

P2P-3(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

P2P-3(config-if)#interface Serial0/1/1
P2P-3(config-if)# ip address 10.0.0.10 255.255.255.252
P2P-3(config-if)# no shutdown

P2P-3(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

P2P-3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

P2P-3(config-if)#interface GigabitEthernet0/0/0
P2P-3(config-if)# ip address 192.168.3.1 255.255.255.240
P2P-3(config-if)# no shutdown

P2P-3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

P2P-3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

P2P-3(config-if)#router ospf 10
P2P-3(config-router)# passive-interface GigabitEthernet0/0/0
P2P-3(config-router)# auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.
P2P-3(config-router)# network 10.0.0.4 0.0.0.3 area 0
P2P-3(config-router)# network 10.0.0.8 0.0.0.3 area 0
P2P-3(config-router)# network 192.168.3.0 0.0.0.15 area 0
P2P-3(config-router)#end
P2P-3#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
P2P-3#
P2P-3#
```

แสดงคำสั่ง config ของ P2P-3 ครบทั้ง IP Address และ OSPF network statement ตามที่ต้องการ พร้อม passive-interface บน G0/0/0

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname BC-1
BC-1(config)#interface Serial0/1/0
BC-1(config-if)# ip address 10.0.0.14 255.255.255.252
BC-1(config-if)# ip ospf hello-interval 20
BC-1(config-if)# ip ospf dead-interval 80
BC-1(config-if)# ip ospf 10 area 0
OSPF process 10 cannot start. There must be at least one "up" IP interface
BC-1(config-if)# no shutdown

BC-1(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

BC-1(config-if)#interface Serial0/1/1
BC-1(config-if)# ip address 64.0.100.2 255.255.255.252
BC-1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
BC-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

BC-1(config-if)#interface GigabitEthernet0/0/0
BC-1(config-if)# ip address 10.0.1.1 255.255.255.248
BC-1(config-if)# ip ospf priority 255
BC-1(config-if)# ip ospf 10 area 0
BC-1(config-if)# no shutdown

BC-1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

BC-1(config-if)#router ospf 10
BC-1(config-router)# router-id 6.6.6.6
BC-1(config-router)# passive-interface Serial0/1/1
BC-1(config-router)# auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
BC-1(config-router)# default-information originate
BC-1(config-router)#
BC-1(config-router)#ip route 0.0.0.0 0.0.0.0 Serial0/1/1
%Default route without gateway, if not a point-to-point interface, may impact performance
BC-1(config)#end
BC-1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
BC-1#

```

แสดงคำสั่ง config ของ BC-1 ได้แก่ Router ID 6.6.6.6, การตั้ง priority 255 บน G0/0/0 (เพื่อให้เป็น DR เสมอ), default-information originate, และ static default route ออกทาง exit-interface (Serial0/1/1) ตาม requirement

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname BC-2
BC-2(config)#interface GigabitEthernet0/0/0
BC-2(config-if)# ip address 192.168.4.1 255.255.255.252
BC-2(config-if)# ip ospf 10 area 0
OSPF process 10 cannot start. There must be at least one "up" IP interface
BC-2(config-if)# no shutdown

BC-2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

BC-2(config-if)#interface GigabitEthernet0/0/1
BC-2(config-if)# ip address 10.0.1.2 255.255.255.248
BC-2(config-if)# ip ospf 10 area 0
BC-2(config-if)# no shutdown

BC-2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

BC-2(config-if)#router ospf 10
BC-2(config-router)# router-id 5.5.5.5
BC-2(config-router)# passive-interface GigabitEthernet0/0/0
BC-2(config-router)# auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
      Please ensure reference bandwidth is consistent across all routers.
BC-2(config-router)#end
BC-2#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
BC-2#
BC-2#
```

แสดงคำสั่ง config ของ BC-2 ได้แก่ Router ID 5.5.5.5, การเปิด OSPF แบบ interface-based (ip ospf 10 area 0) บนทั้ง G0/0/0 และ G0/0/1 และตั้ง passive-interface บนพอร์ต LAN (G0/0/0)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname BC-3
BC-3(config)#interface GigabitEthernet0/0/0
BC-3(config-if)# ip address 192.168.5.1 255.255.255.0
BC-3(config-if)# ip ospf 10 area 0
OSPF process 10 cannot start. There must be at least one "up" IP interface
BC-3(config-if)# no shutdown

BC-3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

BC-3(config-if)#interface GigabitEthernet0/0/1
BC-3(config-if)# ip address 10.0.1.3 255.255.255.248
BC-3(config-if)# ip ospf 10 area 0
BC-3(config-if)# no shutdown

BC-3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

BC-3(config-if)#router ospf 10
BC-3(config-router)# router-id 4.4.4.4
BC-3(config-router)# passive-interface GigabitEthernet0/0/0
BC-3(config-router)# auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.
BC-3(config-router)#end
BC-3#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
BC-3#
BC-3#
```

แสดงคำสั่ง config ของ BC-3 เช่นเดียวกับ BC-2 โดยใช้ Router ID 4.4.4.4

Objectives

Implement single-area OSPFv2 in both point-to-point and broadcast multiaccess networks.

Background

You are helping a network engineer test an OSPF set up by building the network in the lab where you work. You have interconnected the devices and configured the interfaces and have connectivity within the local LANs. Your job is to complete the OSPF configuration according to the requirements left by the engineer.

Use the information provided and the list of requirements to configure the test network. When the task has been successfully completed, all hosts should be able to ping the internet server.

Instructions

Configure the network to meet the requirements.

Requirements

Use process ID 10 for OSPF activation on all routers.

- Activate OSPF using network statements and inverse masks on the routers in the Headquarters network.
- Activate OSPF by configuring the interfaces of the network devices in the Data Service network, where required.
- Configure router IDs on the multiaccess network routers as follows:
 - BC-1: 6.6.6.6
 - BC-2: 5.5.5.5
 - BC-3: 4.4.4.4
- Configure OSPF so that routing updates are not sent into networks where they are not required.
- Configure router BC-1 with the highest OSPF interface priority so that it will always be the designated router of the multiaccess network.
- Configure a default route to the ISP cloud using the exit interface command argument.
- Automatically distribute the default route to all routers in the network.
- Configure the OSPF routers so that the Gigabit Ethernet interface cost will be 10 and the Fast Ethernet cost will be 100.
- Configure the OSPF cost value of P2P-1 interface Serial0/1/1 to 50.
- Configure the hello and dead timer values on the interfaces that connect P2P-1 and BC-1 to be twice the default values.

ตรวจสอบที่ละ Requirement

ทดสอบ Process ID 10 ทุก router

show ip protocols

```
P2P-1#
P2P-1#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.0.0.13
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    10.0.0.8 0.0.0.3 area 0
    10.0.0.12 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.13         110          00:12:36
    192.168.2.1       110          00:13:04
    192.168.3.1       110          00:13:04
  Distance: (default is 110)
```

```
P2P-3>
P2P-3>show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.3.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.4 0.0.0.3 area 0
    10.0.0.8 0.0.0.3 area 0
    192.168.3.0 0.0.0.15 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.13         110          00:14:20
    192.168.2.1       110          00:14:46
    192.168.3.1       110          00:14:46
  Distance: (default is 110)
```

```
Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.2.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    10.0.0.4 0.0.0.3 area 0
    192.168.1.0 0.0.0.255 area 0
    192.168.2.0 0.0.0.255 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
    GigabitEthernet0/0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.13         110          00:13:40
    192.168.2.1       110          00:14:07
    192.168.3.1       110          00:14:07
  Distance: (default is 110)
```

```
BC-1>
BC-1>show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 6.6.6.6
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Passive Interface(s):
    Serial0/1/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    6.6.6.6          110          00:19:05
    10.0.0.14         110          00:44:15
  Distance: (default is 110)
```

BC-1>

```
BC-2>show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 5.5.5.5
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    5.5.5.5          110          00:11:24
    192.168.4.1       110          00:41:31
  Distance: (default is 110)
```

BC-2>

```
BC-3>show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.4
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    4.4.4.4          110          00:10:40
    192.168.5.1       110          00:40:46
  Distance: (default is 110)
```

BC-3>

☒ ต้องเห็นบรรทัด: Routing Protocol is "ospf 10"

OSPF Neighbor

P2P-1

```
P2P-1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
6.6.6.6	0	FULL/ -	00:01:09	10.0.0.14	Serial0/1/1
192.168.3.1	0	FULL/ -	00:00:33	10.0.0.10	Serial0/2/0
192.168.2.1	0	FULL/ -	00:00:37	10.0.0.2	Serial0/1/0

```
P2P-1#
```

ผลการรัน show ip ospf neighbor ที่ P2P-1 แสดงว่ามองเห็น neighbor ครบทั้ง 3 ตัว คือ BC-1 (6.6.6.6), P2P-3 (192.168.3.1) และ P2P-2 (192.168.2.1) ทุกตัวอยู่ในสถานะ **FULL** ยืนยันว่า OSPF adjacency ในโซน Headquarters ทำงานสมบูรณ์

BC-1

```
BC-1#  
BC-1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.0.13	0	FULL/ -	00:01:00	10.0.0.13	Serial0/1/0

```
BC-1#
```

ผลการรัน show ip ospf neighbor ที่ BC-1 แสดงว่าเห็น P2P-1 (10.0.0.13) เป็น **FULL** ผ่าน Serial0/1/0 ยืนยันว่าลิงก์ point-to-point ระหว่าง Headquarters และ Data Services เชื่อมต่อสำเร็จ

ทดสอบ

Router ID

BC-1

```
BC-1#  
BC-1#show ip ospf  
Routing Process "ospf 10" with ID 6.6.6.6  
Supports only single TOS(TOS0) routes  
Supports opaque LSA  
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs  
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs  
Number of external LSA 1. Checksum Sum 0x006454  
Number of opaque AS LSA 0. Checksum Sum 0x000000  
Number of DCbitless external and opaque AS LSA 0  
Number of DoNotAge external and opaque AS LSA 0  
Number of areas in this router is 1. 1 normal 0 stub 0 nssa  
External flood list length 0  
BC-1#S|
```

ผลการรัน show ip ospf แสดง Router ID 6.6.6.6 ตรงตาม requirement และสถานะ Process OSPF 10 ทำงานปกติ

BC-2

```
BC-2>  
BC-2>show ip ospf  
Routing Process "ospf 10" with ID 5.5.5.5  
Supports only single TOS(TOS0) routes  
Supports opaque LSA  
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs  
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs  
Number of external LSA 0. Checksum Sum 0x000000  
Number of opaque AS LSA 0. Checksum Sum 0x000000  
Number of DCbitless external and opaque AS LSA 0  
Number of DoNotAge external and opaque AS LSA 0  
Number of areas in this router is 1. 1 normal 0 stub 0 nssa  
External flood list length 0  
BC-2>|
```

ผลการรัน show ip ospf แสดง Router ID 5.5.5.5 ตรงตาม requirement

BC-3

```
BC-3>  
BC-3>  
BC-3>show ip ospf  
Routing Process "ospf 10" with ID 4.4.4.4  
Supports only single TOS(TOS0) routes  
Supports opaque LSA  
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs  
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs  
Number of external LSA 0. Checksum Sum 0x000000  
Number of opaque AS LSA 0. Checksum Sum 0x000000  
Number of DCbitless external and opaque AS LSA 0  
Number of DoNotAge external and opaque AS LSA 0  
Number of areas in this router is 1. 1 normal 0 stub 0 nssa  
External flood list length 0  
BC-3>
```

ผลการรัน show ip ospf แสดง Router ID 4.4.4.4 ตรงตาม requirement

ทดสอบ Interface Cost

P2P-1

```
P2P-1>
P2P-1>show ip ospf interface Serial0/1/1

Serial0/1/1 is up, line protocol is up
  Internet address is 10.0.0.13/30, Area 0
  Process ID 10, Router ID 10.0.0.13, Network Type POINT-TO-POINT, Cost: 6476
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 20, Dead 80, Wait 80, Retransmit 5
    Hello due in 00:00:18
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 6.6.6.6
  Suppress hello for 0 neighbor(s)
P2P-1>
```

ผลการรัน show ip ospf interface Serial0/1/1 แสดงว่าลิงก์นี้อยู่ใน Area 0, ตั้ง Timer เป็น Hello 20/Dead 80 ตามที่ requirement กำหนด (2 เท่าของค่า default) และมี Adjacent neighbor กับ BC-1 (6.6.6.6) สำเร็จสมบูรณ์

BC-1

```
BC-1#
BC-1#show ip ospf interface GigabitEthernet0/0/1

GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 10.0.1.1/29, Area 0
  Process ID 10, Router ID 6.6.6.6, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 255
  Designated Router (ID) 6.6.6.6, Interface address 10.0.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
```

ผลการรัน show ip ospf interface GigabitEthernet0/0/1 แสดงว่า BC-1 มี Priority 255 และมีสถานะเป็น DR (Designated Router) ของเครือข่าย Multiaccess ตาม requirement ที่กำหนดให้ BC-1 ต้องเป็น DR เสมอ และ Cost ถูกปรับเป็น 10 ตามที่กำหนด (Gigabit Ethernet cost = 10)

ทดสอบ DR Election

```
BC-1#
BC-1#show ip ospf interface GigabitEthernet0/0/1

GigabitEthernet0/0/1 is up, line protocol is up
Internet address is 10.0.1.1/29, Area 0
Process ID 10, Router ID 6.6.6.6, Network Type BROADCAST, Cost: 10
Transmit Delay is 1 sec, State DR, Priority 255
Designated Router (ID) 6.6.6.6, Interface address 10.0.1.1
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:04
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
BC-1#
```

ผลการรัน show ip route | include 0.0.0.0 แสดงว่า BC-1 มี static default route (S* 0.0.0.0/0) ที่ออกจาก Serial0/1/1 (exit-interface) ตาม requirement ที่กำหนดให้ใช้ exit-interface แทน next-hop IP

ทดสอบ Default Route กระจายทั่วเครือข่าย

BC-1

```
BC-1#
BC-1#show ip route | include 0.0.0.0
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
    10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
O        10.0.0.0/30 [110/12952] via 10.0.0.13, 00:32:24, Serial0/1/0
S* 0.0.0.0/0 is directly connected, Serial0/1/1
BC-1#
```

P2P-2

```
P2P-1>
P2P-1>show ip ospf interface Serial0/1/1

Serial0/1/1 is up, line protocol is up
Internet address is 10.0.0.13/30, Area 0
Process ID 10, Router ID 10.0.0.13, Network Type POINT-TO-POINT, Cost: 6476
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 20, Dead 80, Wait 80, Retransmit 5
Hello due in 00:00:18
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 6.6.6.6
Suppress hello for 0 neighbor(s)
P2P-1>
P2P-1>
P2P-1>show ip route | include 0.0.0.0
Gateway of last resort is 10.0.0.14 to network 0.0.0.0
    10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
C        10.0.0.0/30 is directly connected, Serial0/1/0
O*E2 0.0.0.0/0 [110/1] via 10.0.0.14, 00:33:00, Serial0/1/1
P2P-1>
```

ส่วนบนแสดง adjacency กับ BC-1 สมบูรณ์ (FULL) ส่วนล่างแสดงผลการรัน show ip route | include 0.0.0.0 ที่ P2P-1 ซึ่งเห็น default route แบบ O*E2 0.0.0.0/0 ที่เรียนรู้มาจาก BC-1 ผ่าน OSPF (via 10.0.0.14) ยืนยันว่า default route ถูกกระจาย (distributed) มาถึง Headquarters สำเร็จตาม requirement ข้อ "Automatically distribute the default route to all routers"